DS5216: Artificial Intelligence (2024/25) Programming Assignment 02: Player Tracking in Sports Videos Report Document

Performance Comparison of Models

The models were evaluated using the standardized validation benchmarks (coco8 and coco8-pose) to determine performance trends in bounding box object detection and key point tracking.

Performance Metrics Table -

When analyzing your sports tracking pipeline, look specifically at the person class for the detection model (since players are classified as persons) and the Pose metrics for the key point model.

Model Task / Variant	Target Category	Precision (P)	Recall (R)	mAP@50	mAP@50-95
Player Detection (yolov8n.pt)	person class only	0.7210	0.5000	0.5170	0.2650
Player Detection (yolov8n.pt)	All Classes (Average)	0.6210	0.8333	0.8875	0.6291
Key point Detection (yolov8n-pose.pt)	Pose (Key points)	0.8475	0.5000	0.5269	0.3453

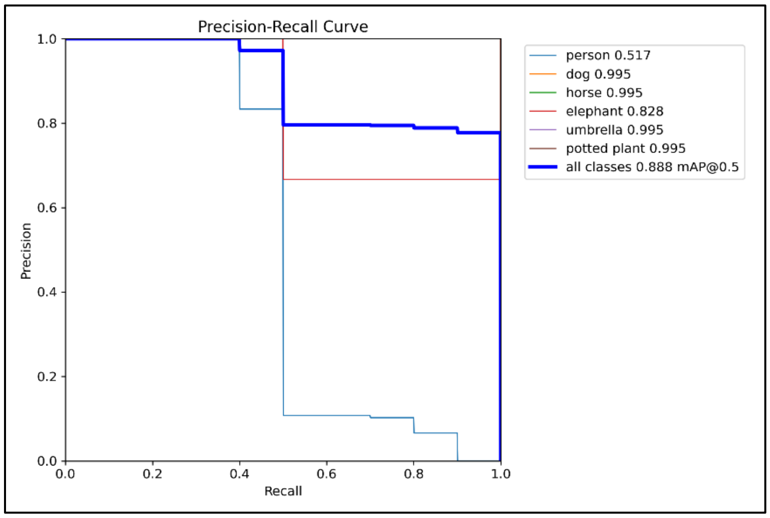
Metrics Interpretation -

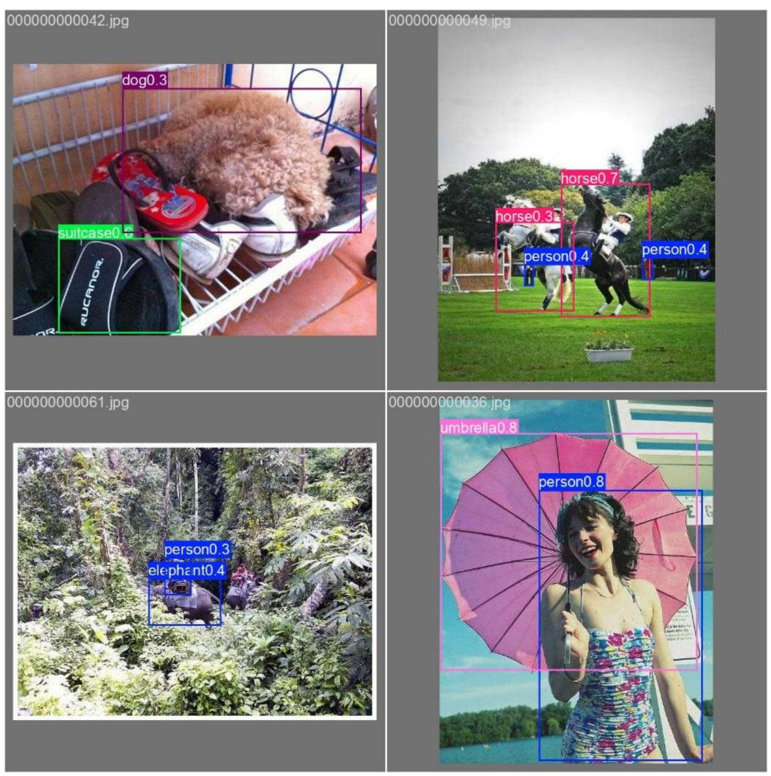
- Precision vs. Recall: The Key point Pose model exhibits a high Precision (0.8475) but a lower Recall (0.5000). This means that when the model maps out a skeleton, it is highly confident and accurate about the joint placements, but it misses about half of the total potential key point configurations in complex or obscured scenes.
- mAP@50 Evaluation: The overall detection framework handles generic shapes aggressively ($mAP50 = 0.8875$), but narrowing it down directly to human targets reveals room for sport-specific optimizations ($mAP50 = 0.5170$).

Loss Curves and Visual Outputs -

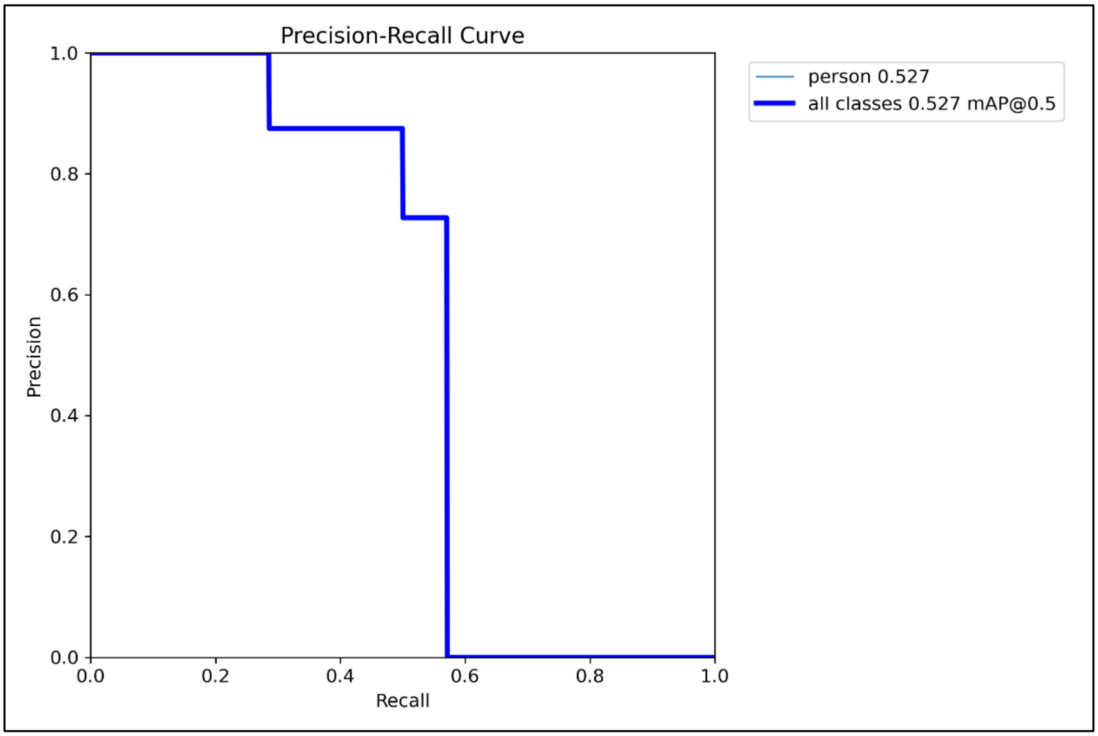
The validation metrics and trend curves were automatically saved to your Colab workspace directory during execution. For your report's visual evidence, the following figures should be retrieved and included from your folders:

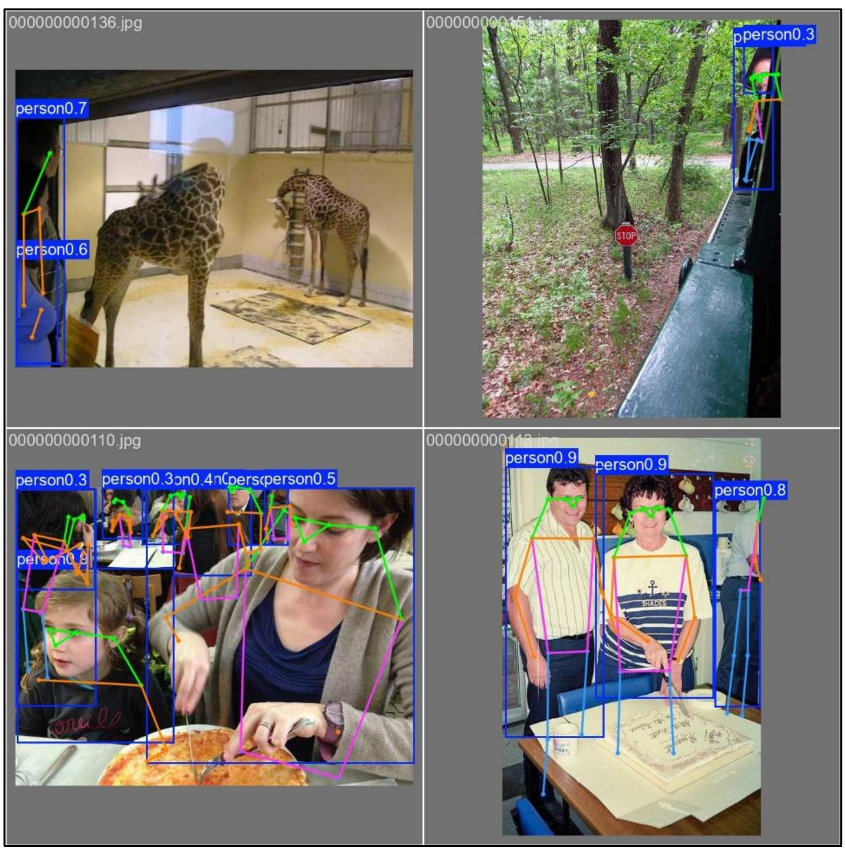
- **From /content/runs/detect/val/:** BoxPR_curve.png (Precision-Recall trade-off curve) and val_batch0_pred.jpg (Visual snapshot of bounding boxes).





- **From /content/runs/pose/val/:** PosePR_curve.png (Key point PR visualization) and val_batch0_pred.jpg (Visual verification of skeleton overlays).





<u>Discussion on Model Performance, Limitations, and Improvements</u>

The baseline evaluation highlights that both the detection and pose variants of YOLOv8-Nano are exceptionally lightweight, featuring approximately 3.15M to 3.29M parameters and utilizing under 10 GFLOPs of computational power.

This hyper-efficiency results in incredibly fast inference speeds (~325ms to 395ms on a basic shared cloud CPU), making this specific setup optimal for embedding into continuous sports tracking architectures (like tracking players live with ByteTrack) without requiring expensive GPU setups.

Limitations in Sports Contexts -

1. Low Recall under Motion (The 0.50 Recall Boundary): A recall score of 0.5000 indicates that the model struggles to identify targets when they are partially occluded or in atypical orientations. In actual sports clips, players routinely overlap (e.g., defenders blocking an attacker), which leads to temporary tracking loss or dropped pose detections.
2. Fast Action Blur: The pretrained models are evaluated on static, crisp benchmark frames. High-velocity actions common in sports (e.g., throwing a football, swinging a cricket bat) produce motion blur that heavily degrades key point spatial alignment.
3. Scale Variance: Far-away stadium camera feeds reduce players to minor clusters of pixels. Given YOLOv8-Nano's compressed resolution design, small bounding boxes are often discarded entirely or misclassified.

Possible Improvements -

- Sport-Specific Fine-Tuning: The pretrained model spent processing time validating irrelevant classes like dog, horse, or potted plant. Re-training or fine-tuning the architecture exclusively on a sports tracking dataset (e.g., *Soccer Net* or *SportsMOT*) would refocus all parameters purely onto athletes and sports equipment.
- Upgrading Architecture Scale: Transitioning from the Nano (yolov8n) model to the Small (yolov8s) or Medium (yolov8m) variant will marginally increase processing latency but heavily improve both the Box and Pose Recall boundaries.
- Deep Spatial-Temporal Trackers: Supplementing ByteTrack with a Re-Identification (Re-ID) tracking framework would allow the pipeline to remember player tracks based on visual features (like uniform color/jersey text) even when the pose skeleton is lost during a huddle or a crash.